

PRODUCT CONCEPT

The Household Financial Interchange

A bench of specialist agents, one shared ledger, and a human gate on everything that moves money.

Assets · Taxes · Strategy · Markets · Estate



THE PROBLEM

One financial life, a dozen systems, no shared memory

6–12

institutions, brokerages and
wallets holding pieces of it

4+

tax deadlines a year that
arrive without a reminder

1

person doing every
reconciliation by hand

0

places where the whole
picture actually lives



Every question costs a data-gathering project Answering "can I afford this?" means opening nine tabs and trusting stale balances.



Decisions and deadlines live in your head Why you bought it, what you promised the accountant, when the estimate is due — all untraceable.



Advice arrives with no memory and no receipts A recommendation you cannot audit six months later is a guess with good grammar.

THE CONCEPT

A hub, not a chatbot

Three commitments that make the difference between a demo and something you would trust with a wire transfer.



One source of truth

Agents do not exchange opinions about your balances. They read and write a single dated ledger, and every fact carries where it came from and when it was known.



Specialists, not a generalist

A narrow agent with a narrow tool set is testable. "A financial assistant" is not. Each agent owns one domain and answers for it.



Nothing moves without you

Advisory agents can only propose. Execution sits behind an approval gate and an audit trail that outlives the conversation.

The four agents you named all live here — alongside the unglamorous ones that keep them from being wrong in expensive ways.

Five rules that decide the architecture



Separation of duties

The agents that give advice never hold credentials and never write to the ledger. Borrowed straight from how accounting departments avoid fraud.

01



The ledger is the only truth

Agents coordinate through shared state, not by messaging each other. If two agents disagree, the ledger settles it.

02



Append-only and as-of dated

Corrected 1099s and restated balances arrive as new facts, never overwrites. You can always ask what was known on a given date.

03



Deterministic control flow

The Hub decides what runs next. A model decides what to say inside a step — never who goes next or whether to skip the check.

04



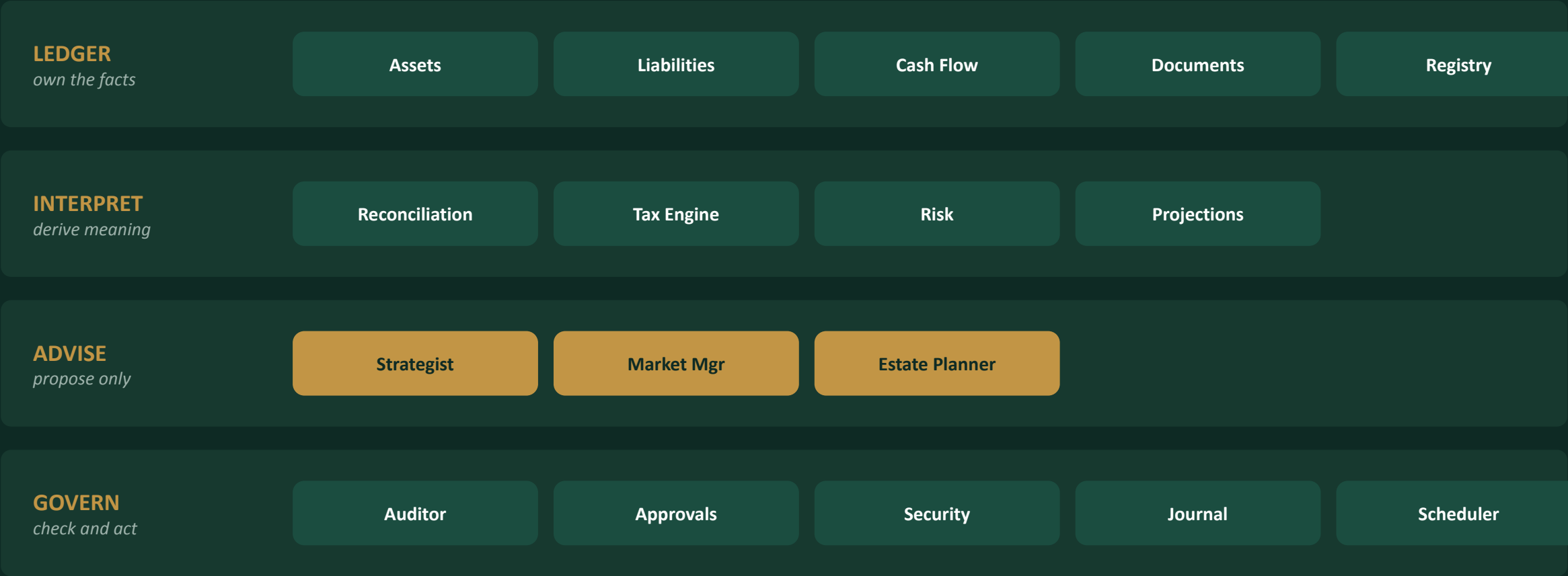
The human is a node, not an audience

Approval is a typed message with a schema, an expiry and a signature. Not a chat prompt you scroll past.

05

Four tiers, seventeen agents, one direction of trust

Facts flow up. Proposals flow down through a gate. Nothing skips a tier.



Gold = the agents that talk. They are the only tier with no write access and no credentials.

Agents that own facts

These are the only agents allowed to write. Each owns one table and defends it.



Assets Manager

Connects to banks, brokerages and Coinbase.
Owns positions, balances, cost basis and tax lots
— with a read-only token per institution.



Liabilities & Obligations

Mortgages, margin, loans, credit lines, recurring
premiums. Net worth is meaningless without this
half, and refi triggers live here.



Cash Flow

Income and spending, classified. Nothing else can
size a tax estimate or answer "can I afford it"
without a flow picture.



Document Vault

Statements, 1099s, K-1s, deeds, policies, trust
instruments. Extracts structured facts and keeps
the original page as evidence.



Entity & Estate Registry

Who owns what, titled how, with which
beneficiary. Trusts, LLCs, joint accounts,
executors, and where the keys physically are.

*Rule: one writer per fact.
No second agent may
correct the ledger behind
the owner's back.*

Agents that turn facts into meaning

Read-only over the ledger. They compute; they never fetch and never trade.



Reconciliation

Matches every institution against the ledger, kills duplicates, and recognises that a transfer between your own accounts is not income.



Tax Engine

Running estimate of what is owed, lot-level gain tracking, wash-sale watch, withholding vs. estimate coverage, and every deadline.



Risk & Insurance

Concentration, FDIC and SIPC limits, custody single points of failure, coverage gaps, and what one bad month actually does to you.



Projections

Scenario and Monte Carlo engine. Retirement, a property sale, a bad decade. Deterministic maths in code, narrated by a model.

Arithmetic never happens in a language model. Every figure on this tier is produced by code the auditor can re-run and reproduce exactly.

Agents that talk — and can only propose

These three get context and reasoning — and no keys at all.



Strategist

The chat surface. Brainstorms structure, trade-offs, and long-horizon questions across finances, wills and estate. Reads aggregates; writes only to the decision journal.



Market Manager

Candidate positions, drift against target allocation, harvest opportunities. Every idea emitted as a proposal with a thesis, an evidence list and an expiry date.



Estate Planner

Compares the plan on paper to the plan in reality: titling, beneficiaries, document versions, digital access, and who can operate this system when you cannot.

Three constraints on everything this tier produces

Cite or stay quiet

Every claim links to ledger facts by id. No evidence, no proposal.

Expire by default

A recommendation older than its window is dead, not stale advice.

Log the reasoning

The thesis is stored so you can grade it against reality next year.

Agents that check, gate, and remember

The tier that turns a clever assistant into something with an audit trail.



Auditor

Reviews every proposal before you see it — plan conflicts, wash-sale risk, tax-lot impact, and any figure it cannot reproduce.



Approvals & Execution

The only agent with write access to the outside world. Holds the gate, executes what you signed, captures the receipt, reconciles the fill.



Security & Access

Credential custody, token rotation, scope enforcement, and an access log of which agent read what and when.



Decision Journal

What you decided, when, why, and what you expected. Financial feedback loops are years long; memory is the only way to learn from them.



Scheduler & Tickler

Owens time. Fires the recurring workflows, escalates deadlines, and chases the things that quietly did not happen.

If you build only one agent from this tier, build the Auditor.

Reconciliation is the agent that decides whether any of the rest is true

Pull from six institutions and the data will disagree with itself within a week. Every downstream agent — tax, strategy, markets — inherits that error and states it with confidence.

Somebody has to own the question "is this actually right?", and it cannot be the same agent that fetched the data.

1 A transfer between your own accounts booked twice as income

2 A balance that stopped updating four days ago but still looks live

3 A corrected 1099 that silently changes last year's answer

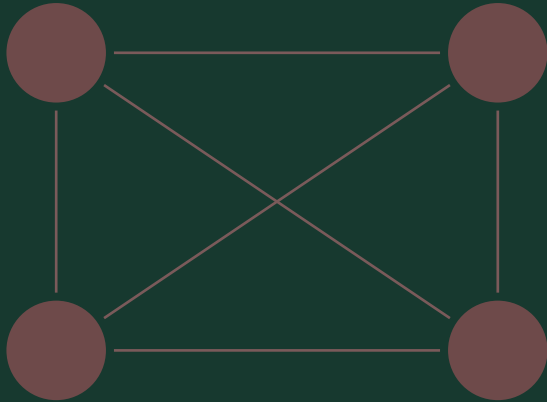
4 Cost basis missing on transferred lots, quietly assumed as zero

5 A crypto swap that is a taxable event no bank feed will mention

Each of these is silent, plausible, and expensive. All five are cheap to detect if one agent is looking for them.

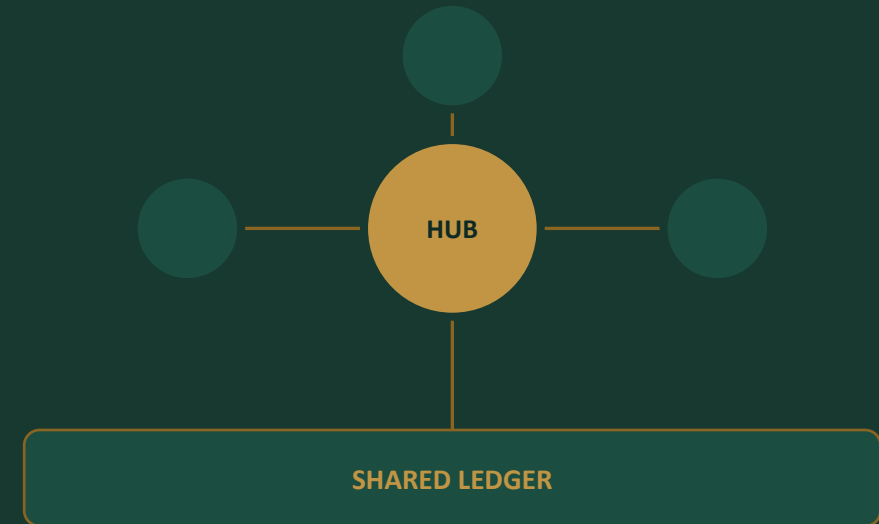
Agents should not chat with each other

TOO MUCH COMMUNICATION FAILS



Every agent messages every other agent. Errors compound, cost is unbounded, and no one can say why a number changed.

WHAT SURVIVES CONTACT WITH MONEY



Agents subscribe to events and read shared state. The Hub owns sequencing; the ledger owns truth.

Practically: a blackboard for state, an event bus for triggers, and a deterministic state machine for anything that touches money.

Six message types, every one of them typed

Recommendation

```
id:          rec_2026-08-19-014
from:        market_manager
subject:     acct.brokerage.taxable
action:      SELL 40 VTI
thesis:      "equity 6.2% over target"
evidence:    [pos_881, plan_target_v3]
as_of:       2026-08-19T22:04Z
tax_lots:    [lot_44 LTCG, lot_51 STCG]
confidence:  0.71
requires:    [auditor.review,
              human.approve]
expires:     2026-08-26
```

Fact A dated observation with a source. Only ledger agents emit these.

Finding An interpretation — a break, a risk, a gap. Carries severity.

Recommendation A proposed action with a thesis, evidence and an expiry.

Approval Your signature on a specific proposal id. Scoped and time-boxed.

Instruction What Execution is permitted to do. Derived only from an Approval.

Receipt What actually happened, fed straight back to Reconciliation.

Every message carries provenance, an as-of timestamp and an id. Anything without them is not admissible.

Who is allowed to do what

Capability scoping is enforced by the Hub, not by asking the agent nicely in a prompt.

	Institution credentials	Read ledger	Write ledger	Place orders	Sees full PII
Assets Manager	R/O token	yes	own tables	no	yes
Tax Engine	no	yes	own tables	no	yes
Strategist	no	aggregates	journal only	no	masked
Market Manager	no	positions	no	no	masked
Auditor	no	yes	no	no	masked
Approvals & Execution	scoped, per-order	yes	receipts	on approval	yes

The Market Manager never learns your account numbers, and the Strategist never sees a balance it does not need. **A compromised advisory agent should be embarrassing, not catastrophic.**

What the Hub actually runs, and how often

Continuous

- Feed ingestion & normalisation
- Anomaly and break detection
- Document intake and extraction

Daily

- Reconcile all institutions
- Cash position & liquidity check
- Exception queue triage

Monthly

- Net worth close & snapshot
- Spend vs. plan review
- Drift check against targets

Quarterly

- Estimated tax calculation & payment
- Rebalance proposal cycle
- Access & credential review

Annual

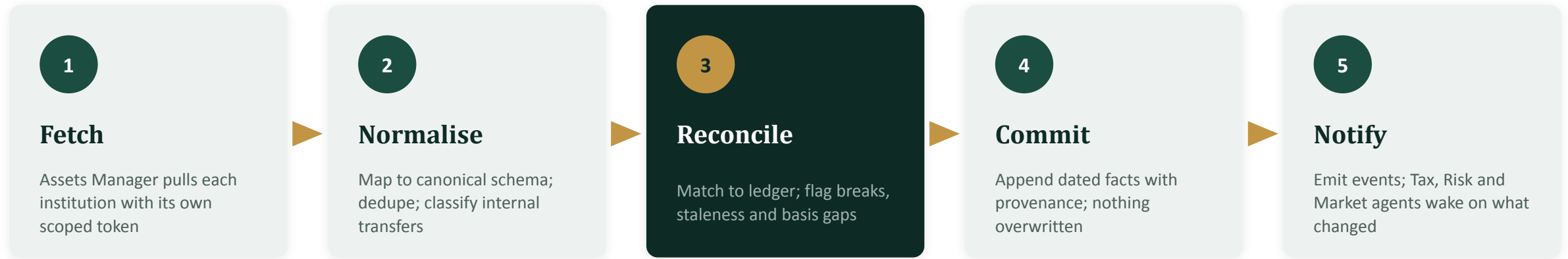
- Year-end tax planning window
- Tax-loss harvest sweep
- Estate & beneficiary hygiene audit
- Insurance and coverage review

Event-driven

- Life events: sale, move, inheritance
- New account onboarding
- Proposal → approval → execution

Nightly: ingest, normalise, reconcile

The workflow that runs while you sleep, and the one that must never fail silently.



The exception path is the real product

Break found

Ledger marked provisional for that account, not silently patched

Downstream held

Tax and Market agents refuse to run on provisional data

Queued for you

One item in the morning queue, with both versions side by side

Resolved as a fact

Your answer is appended and dated — the history stays intact

A reconciliation agent that resolves ambiguity on its own is worse than none at all.

Proposal → review → approval → execution

The path any action must walk. Six steps, one of them irreducibly human.



the only irreducibly human step

What the Auditor blocks on

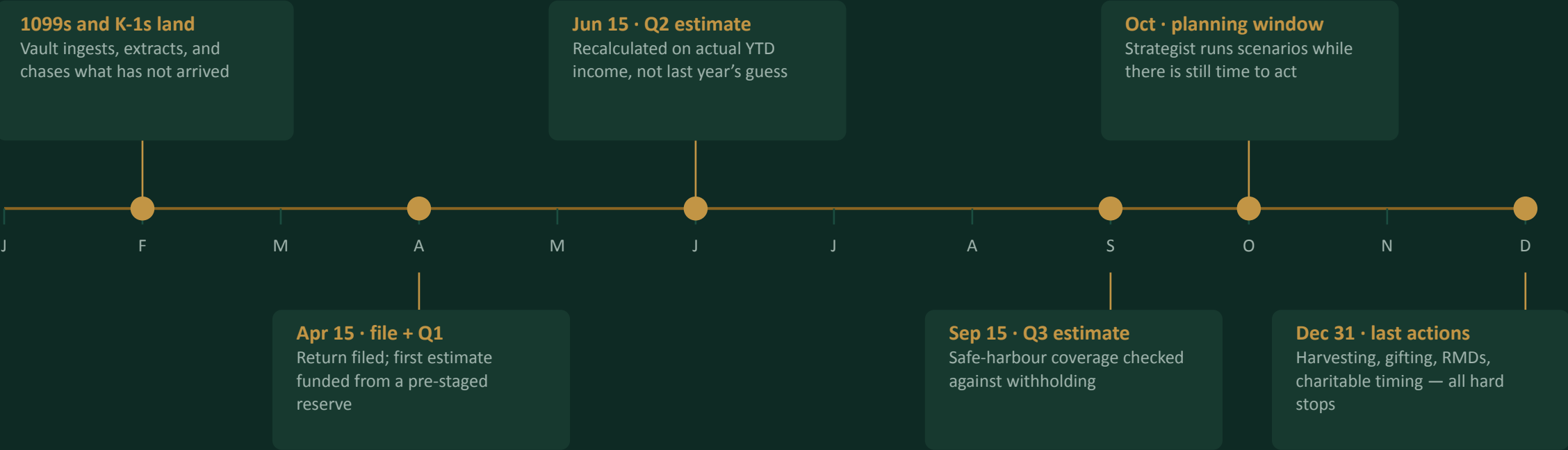
- A figure it cannot reproduce from ledger facts
- Wash-sale collision, or a lot choice that quietly costs you
- Conflict with a standing constraint or the written plan
- Cash needed for a tax payment inside the horizon

What your approval actually is

- Scoped to one proposal id, not a standing permission
- Bounded — max size, limit price, expiry attached
- Revocable until the instruction is actually sent
- Stored with the thesis you approved, for later review

The tax year, as a standing workflow

Your Taxes agent stops being a reminder service and becomes a calendar with a balance sheet behind it.



One event, eight consequences

Life events are where a connected system pays for itself — and where a pile of separate apps fails.

TRIGGER

You sell the rental property

A single fact enters the ledger. The Hub fans it out to every agent that holds a claim on it, and each one answers for its own domain.

▶ **Tax Engine** Recomputes the gain, depreciation recapture and the Q3 estimate

▶ **Cash Flow** Reserves the tax before the proceeds get spent or reinvested

▶ **Liabilities** Closes the mortgage, ends the escrow and the insurance premium

▶ **Risk** Flags the new cash concentration and the lapsed liability coverage

▶ **Market Manager** Proposes a staged deployment, not a lump-sum decision

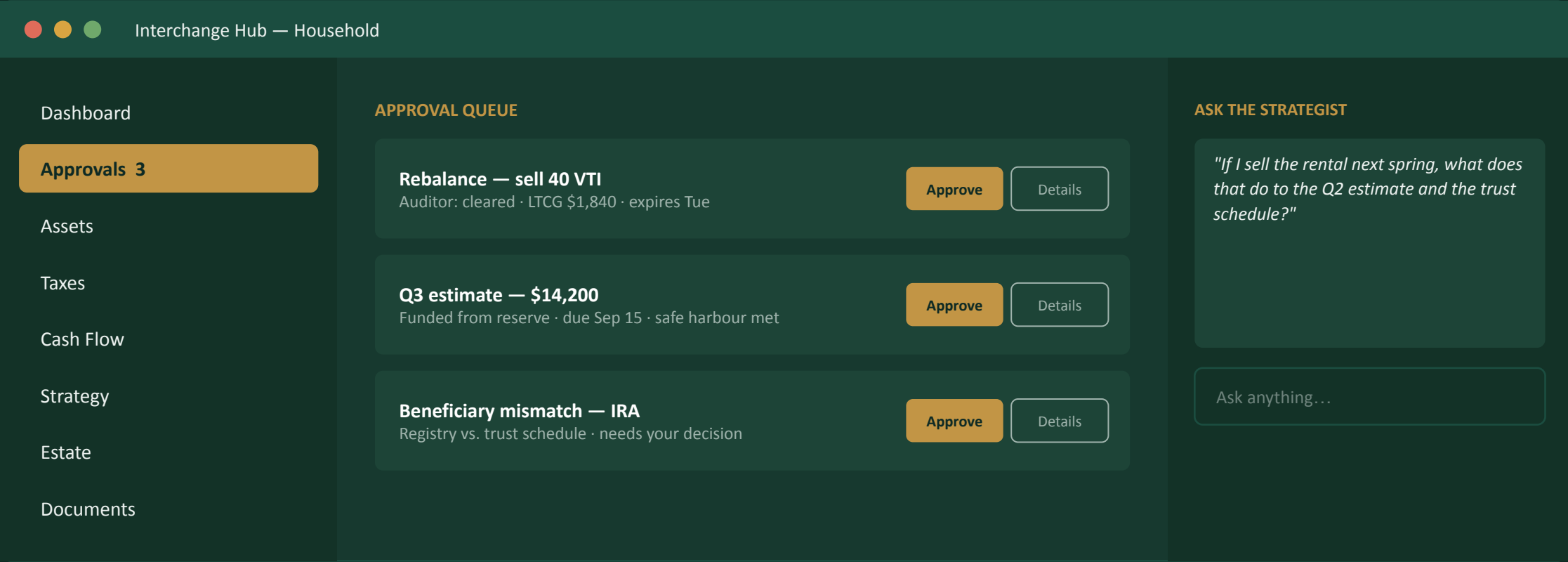
▶ **Estate Planner** Removes the deed from the trust schedule; flags the will clause

▶ **Registry** Retires the entity, archives the title chain and the closing file

▶ **Journal** Records why you sold, and what you expected the proceeds to do

Every one of these is a sub-workflow with its own approval step. None of them depends on you remembering it.

Workflows on the left, evidence in the middle, decisions on the right



One rule for the GUI: every number is clickable back to the fact, the source document, and the date it was observed.

Build it in the order that trust accumulates

Each phase is useful on its own, and each one is a precondition for the next.

01

Ledger first

- Assets, Liabilities, Cash Flow
- Document Vault + extraction
- Reconciliation, read-only

Boring, and everything else rests on it.

02

Time and obligations

- Tax Engine + Scheduler
- Deadline and payment workflows
- Exception queue in the GUI

First real payoff: nothing surprises you.

03

Judgement

- Strategist chat
- Projections and scenarios
- Estate registry + hygiene audit

Advice that finally has a memory.

04

Action

- Market Manager proposals
- Auditor
- Approvals and execution

Only once the audit trail is real.

The tempting order is the reverse. Market advice demos beautifully on day one and is worth almost nothing without a ledger underneath it that you believe.

What could go wrong, and what it implies



Credential blast radius

Read-only tokens wherever possible, no withdrawal scope on any key, and hardware-backed custody for anything that can move value.



Models and arithmetic

Never let a language model produce a figure. It selects, explains, and drafts; deterministic code computes and the Auditor re-runs it.



Regulatory footing

Automated trading, advice framing and recordkeeping all carry rules. Worth a conversation with your accountant and attorney before phase 4.



Stale data wearing fresh clothes

Feeds break quietly and aggregators change without notice. Every fact needs an observed-at date and a visible staleness threshold.



The gate you stop reading

If the approval queue is long and boring, you will rubber-stamp it. Batch the routine, escalate only what genuinely differs.



Who operates it when you cannot

An estate system your executor cannot read or shut down is a liability. Build the break-glass path and the plain-language export from day one.

Start with the ledger, not the advice.

1

Write the schema before the agents

Position, Transaction, Obligation, Document, Recommendation, Approval. The data model is the architecture.

2

Connect one institution end to end

Fetch, normalise, reconcile, commit, display. One vertical slice tells you more than six half-built agents.

3

Decide the trust boundary early

Which agents ever hold a credential, and whether execution is in scope at all. It is much harder to retrofit.

